Anomaly-Preserving Thermal Image Denoising for Early Peripheral Neuropathy Risk Assessment

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***Abstract*—Diabetic peripheral neuropathy is a common complication of diabetes that can increase the risk of diabetic foot problems. Infrared thermography provides a non-invasive method for analyzing plantar temperature patterns and detecting abnormal thermal regions. However, thermal images may contain noise and acquisition-related distortions that can affect automated analysis. Conventional denoising methods may improve image quality but can also remove small thermal anomalies that are important for assessment. This work proposes an anomaly-preserving thermal image denoising approach that reduces unwanted noise while retaining important thermal patterns and left-right temperature asymmetry. The enhanced images can subsequently be analyzed using machine-learning and deep-learning methods for risk-oriented assessment of diabetic foot complications. The study aims to connect thermal image enhancement with reliable AI-based analysis while evaluating both image quality and preservation of meaningful thermal information.**

***Index Terms*—diabetic peripheral neuropathy, diabetic foot, infrared thermography, thermal imaging, image denoising, anomaly preservation, thermal asymmetry, deep learning**

I. Introduction

Diabetes mellitus can cause several long-term complications, including peripheral neuropathy and diabetic-foot problems. Peripheral neuropathy may reduce sensation in the feet, allowing minor injuries to remain unnoticed and potentially develop into serious complications. Early identification of abnormal foot conditions is therefore important.

Infrared thermography is a non-invasive technique that captures temperature distributions across the body. In plantar thermography, abnormal hotspots, temperature variations, and left-right thermal asymmetry can provide useful information for diabetic-foot and neuropathic-foot assessment. Previous studies have demonstrated the effectiveness of machine learning and deep learning for analyzing plantar thermal images [1]–[3].

However, thermal images can contain sensor noise, low contrast, background interference, and acquisition-related distortions. Although denoising can improve image quality, excessive smoothing may remove localized thermal abnormalities that are important for medical analysis. Recent infrared-image denoising methods have used adversarial learning, feature fusion, and transformer-based approaches [8]–[10], but these methods generally focus on generic infrared images rather than medically important thermal anomalies.

Therefore, this project proposes an **Anomaly-Preserving Thermal Image Denoising** framework for early peripheral neuropathy risk assessment. The objective is to reduce unwanted noise while preserving thermal anomalies and temperature asymmetry for subsequent AI-based analysis.

II. Literature Review

Recent research shows that plantar thermal images can be effectively used for diabetic-foot analysis. Khandakar *et al.* used the Thermal Change Index with machine-learning techniques for diabetic-foot thermogram classification, demonstrating the importance of temperature-distribution features [1]. Muralidhara *et al.* investigated deep-learning-based classification and grading of diabetic-foot ulcerations using plantar thermal images [2]. Munadi *et al.* proposed a deep-learning approach using decision fusion for early diabetic-foot detection [3].

Other studies explored thermal-image features and automated ulcer detection. Alshayeji *et al.* used Bag-of-Features techniques for early diabetic-foot-ulcer detection [4], while Khosa *et al.* investigated multi-level thermographic data with machine-learning and deep-learning models [5]. Feature-selection methods have also been studied for improving classification using diabetic-foot thermograms [6].

Thermal asymmetry is particularly relevant to the proposed work. A deep neural network has been investigated for identifying neuropathic feet from plantar thermographic images, showing the potential of thermal patterns for neuropathy-related assessment [7]. Digital infrared thermography com-

bined with machine learning has also been used to distinguish diabetic and control subjects using thermal features [11].

In parallel, infrared-image denoising research has developed methods based on adversarial transfer learning, feature fusion, and transformer architectures. Guo *et al.* proposed a lightweight adversarial-transfer-learning method for infrared image denoising [8]. Wu *et al.* introduced multi-head feature fusion and detail enhancement for thermal-image denoising [9], while SwinDenoising used local and global feature fusion with a Swin Transformer [10].

Overall, existing studies demonstrate the usefulness of thermal imaging for diabetic-foot analysis and the effectiveness of modern infrared denoising methods. However, diabetic-foot studies mainly focus on classification or detection, while generic denoising methods do not explicitly preserve medically meaningful thermal anomalies. This motivates the proposed combination of thermal denoising, anomaly preservation, and AI-based risk assessment.

REFERENCES

[1] A. Khandakar *et al.*, "Thermal change index-based diabetic foot thermogram image classification using machine learning techniques," *Sensors*, vol. 22, no. 5, Art. no. 1793, 2022.

[2] S. Muralidhara *et al.*, "Holistic multi-class classification and grading of diabetic foot ulcerations from plantar thermal images using deep learning," *Health Information Science and Systems*, vol. 10, Art. no. 21, 2022.

[3] K. Munadi *et al.*, "A deep learning method for early detection of diabetic foot using decision fusion and thermal images," *Applied Sciences*, vol. 12, no. 15, Art. no. 7524, 2022.

[4] M. H. Alshayeji *et al.*, "Early detection of diabetic foot ulcers from thermal images using the bag of features technique," *Biomedical Signal Processing and Control*, vol. 79, Art. no. 104143, 2023.

[5] I. Khosa *et al.*, "Automatic diabetic foot ulcer recognition using multi-level thermographic image data," *Diagnostics*, vol. 13, no. 16, Art. no. 2637, 2023.

[6] A. Hernandez-Guedes *et al.*, "Feature ranking by variational dropout for classification using thermograms from diabetic foot ulcers," *Sensors*, vol. 23, 2023.

[7] "Deep neural net for identification of neuropathic foot in subjects with type 2 diabetes mellitus using plantar foot thermographic images," *Biomedical Signal Processing and Control*, 2024.

[8] W. Guo, Y. Fan, and G. Zhang, "Lightweight infrared image denoising method based on adversarial transfer learning," *Sensors*, vol. 24, no. 20, Art. no. 6677, 2024.

[9] H. Wu *et al.*, "Mini-infrared thermal imaging system image denoising with multi-head feature fusion and detail enhancement network," *Optics and Laser Technology*, vol. 179, 2024.

[10] "SwinDenoising: A local and global feature fusion algorithm for infrared image denoising," *Mathematics*, vol. 12, no. 19, Art. no. 2968, 2024.

[11] "Digital infrared thermography and machine learning for diabetic foot assessment: Thermal patterns and classification," *Journal of Diabetes and Metabolic Disorders*, 2024.

[12] "The use of AI-powered thermography to detect early plantar thermal abnormalities in patients with diabetes," *JMIR Diabetes*, 2025.

[13] P.-C. Lin *et al.*, "Machine learning for diabetic foot care: Accuracy trends and emerging directions in healthcare AI," *Frontiers in Public Health*, 2025.